

Science Question Paper - English Questions Only

Page 3

[]

31(B) Page 3 of 27 P.T.O.

General Instructions :

Read the following instructions carefully and follow them :

- (i) This question paper contain 39 questions. All questions are compulsory.
- (ii) The question paper is divided into three sections — A, B and C.
Section A : Biology (30 marks)
Section B : Chemistry (25 marks)
Section C : Physics (25 marks)
- (iii) The question paper has MCQs, VSAs, SAs, LAs and CBQs. Marks are given against each question.
- (iv) There are case based questions (CBQs) with three sub-questions and are of 4 marks each.
- (v) Divide your answer sheet into three sections as per the question paper –
Section A (Biology), Section B (Chemistry) and Section C (Physics).
It is compulsory to answer each question in its respective section. Do not mix answers of one section into the other section.
- (vi) Instructions are given with each section and question, wherever necessary.
- (vii) Kindly note that a separate question paper has been provided for visually impaired candidates.
- (viii) There is no overall choice in the question paper. However, an internal choice has been provided in few questions. Only one of the choices in such questions must be attempted.

SECTION A

BIOLOGY

1. Select the gland that secretes digestive enzymes as well as hormones. 1

(A) Thyroid

(B) Pituitary

(C) Pancreas

(D) Adrenal

Page 13

[]

31(B) Page 13 of 27 P.T.O.

16. Attempt either subpart (a) or (b) :

(a) (i) Sugarcane and roses are mostly grown by vegetative propagation. Define this mode of reproduction.

(ii) List two advantages of vegetative propagation.

(iii) Distinguish between unisexual and bisexual flowers. Give one example of each kind. 5

OR

(b) (i) Differentiate between self -pollination and cross-pollination. (two points)

(ii) Explain how does pollination lead to fertilisation in flowering plants.

(iii) Name the end -products of sexual reproduction in plants. 5

SECTION B

CHEMISTRY

17. Which of the following is not an exothermic reaction ? 1

(A) Decomposition of vegetable matter into compost

(B) Reaction of water with quick lime

(C) Respiration process taking place in the cells of our body

(D) Decomposition of calcium carbonate to calcium oxide and carbon dioxide

18. Electrolysis of water is an example of : 1

(A) thermal decomposition reaction

(B) photochemical decomposition reaction

(C) double-displacement reaction

(D) electrolytic decomposition reaction

Page 21

[]

31(B) Page 21 of 27 P.T.O.

SECTION C

PHYSICS

30. Work done in displacing an electron between two points having potential difference of 1 V, is : 1

(A) -1.6×10^{-19} C (B) -1.6×10^{-19} J

(C) $+1.6 \times 10^{-19}$ J (D) -1.6×10^{-19} V

31. Angle of deviation in refraction of light through a glass prism is the : 1

(A) Angle between the normal at the face on the emerging side of the prism and the emergent ray.

(B) Angle between the normal at the face on the emerging side of the prism and the incident ray.

(C) Angle between the incident ray and the refracted ray.

(D) Angle between the directions of incident ray and emergent ray.

For Question number 32, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R).

Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

32. Assertion (A) : When light passes from one medium to another medium, the speed of light may increase or decrease.

Reason (R) : Refractive index of one medium with respect to the other medium may be greater or less than 1. 1